4.0 Mini-Intro to Roles/Features

What are roles and features?



Role

Role is a type of server that a computer can assume. These include

- Web roles
- File server roles
- Communication server roles
- Management roles

and more

Feature

An additional, smaller capability that you can install. These include:

- Bitlocker encryption
- Telnet Server and Client
- RIP Listener
- Powershell

Unlike a role, a feature does not drastically change the functionality of a computer within a network.

What should I be concerned with?

Computers should only have roles/features that they need installed.

Examples:

Standard workstations should not have web server roles installed, as that exposes the system to the public.

A computer that doesn’t use telnet should remove telnet, since keeping it just means another method for hackers to compromise the system.

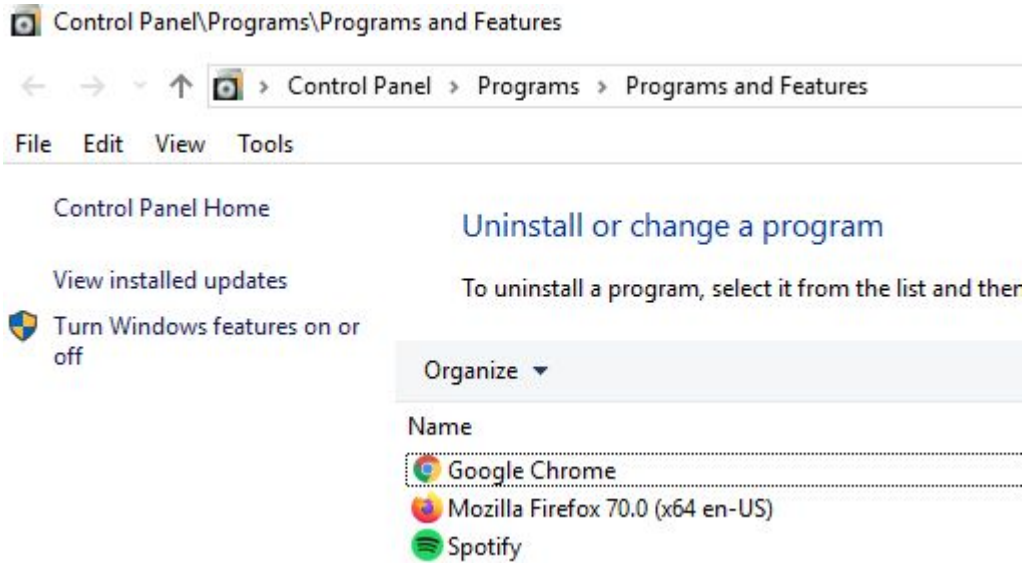
How to manage them?

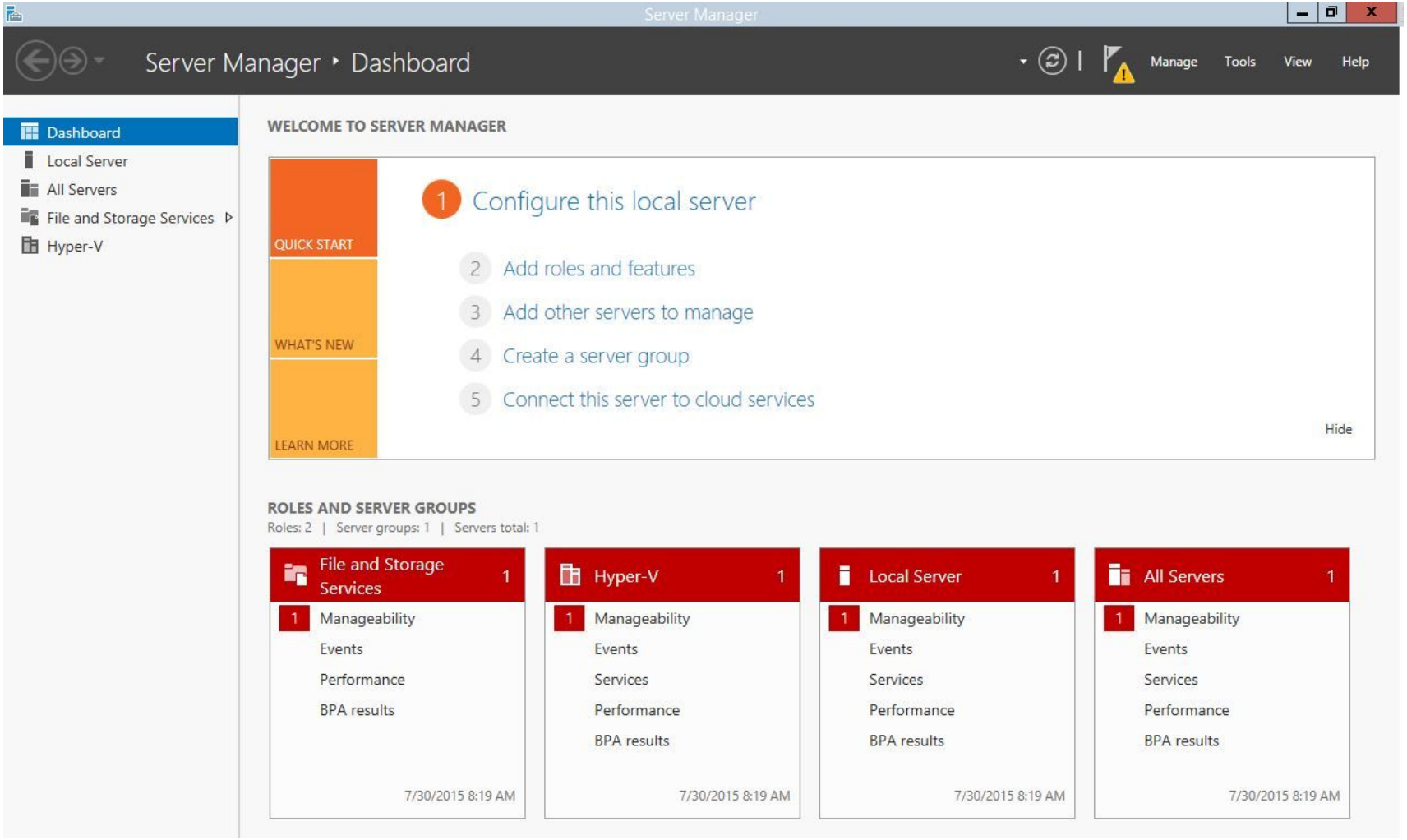
Windows non-server versions (including 10)

- appwiz.cpl

Windows server versions (including 2016 and 2019)

- Use previous appwiz.cpl method OR
- Use server manager, go under “Manage” and look for the “Add/Remove Roles and Features”
    - To find server manager, you can use windows search


Control Panel\Programs\Programs and Features
Control Panel › Programs › Programs and Features
File Edit View Tools
Control Panel Home
View installed updates
Turn Windows features on or off
Uninstall or change a program
To uninstall a program, select it from the list and ther
Organize
Name
Google Chrome
Mozilla Firefox 70.0 (x64 en-US)
Spotify


Server Manager
Server Manager ▸ Dashboard
Manage
Tools
View
Help
Dashboard
Local Server
All Servers
File and Storage Services
Hyper-V
WELCOME TO SERVER MANAGER
QUICK START
WHAT'S NEW
LEARN MORE
1 Configure this local server
2 Add roles and features
3 Add other servers to manage
4 Create a server group
5 Connect this server to cloud services
Hide
ROLES AND SERVER GROUPS
Roles: 2 | Server groups: 1 | Servers total: 1
File and Storage Services 1
1 Manageability
Events
Performance
BPA results
7/30/2015 8:19 AM
Hyper-V 1
1 Manageability
Events
Services
Performance
BPA results
7/30/2015 8:19 AM
Local Server 1
1 Manageability
Events
Services
Performance
BPA results
7/30/2015 8:19 AM
All Servers 1
1 Manageability
Events
Services
Performance
BPA results
7/30/2015 8:19 AM

Role Management (Server)

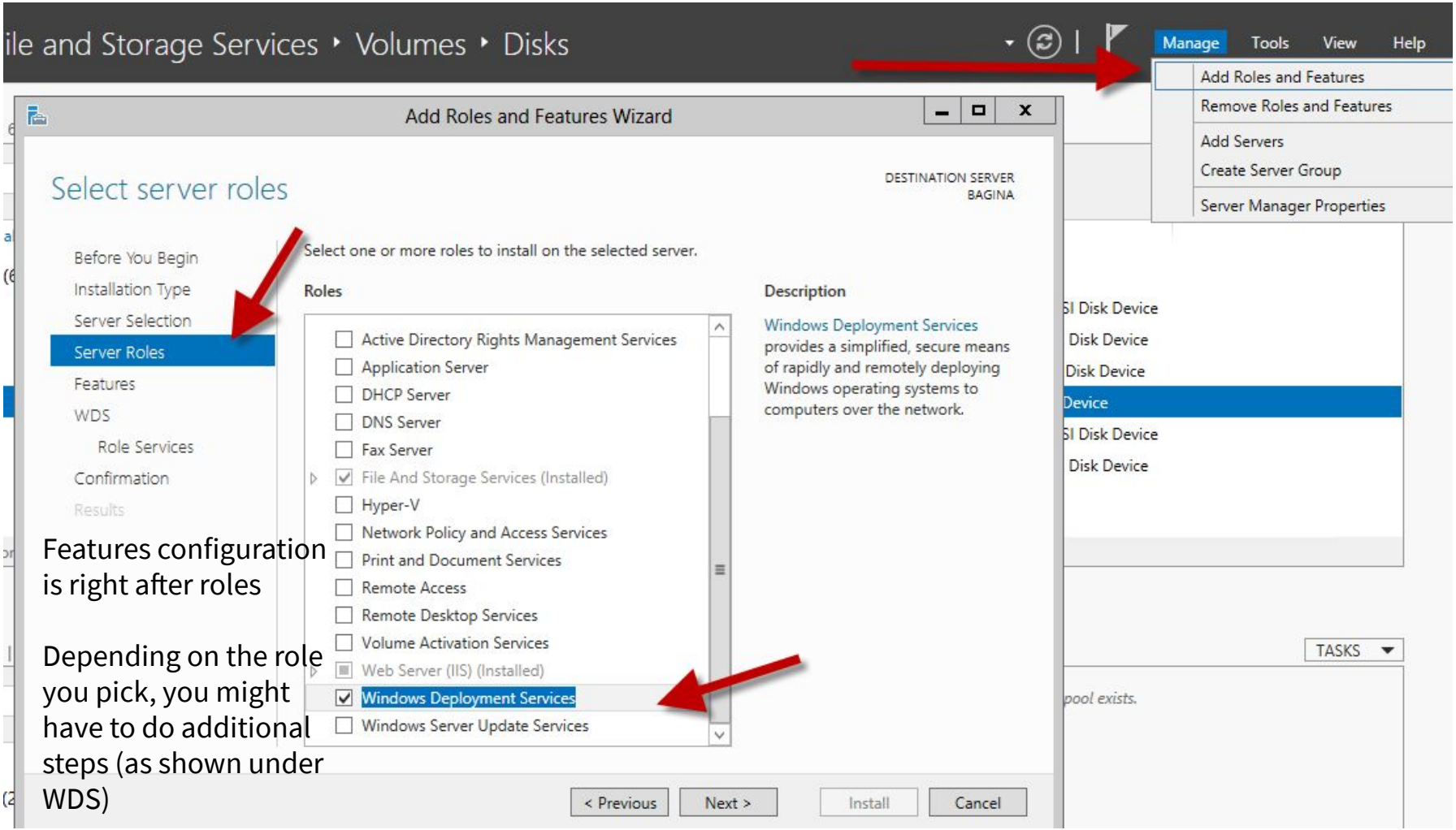
Role management in server versions is a tad bit fancier

Process

- Select role-based or feature-based installation
- Select computer you would like to manage (your own)
- Choose the roles/features you want added/removed

There are two different windows for adding and removing roles/features

First two steps of the process outlined above are automatically set for you, so you just need to click the Next button



Features configuration is right after roles

Depending on the role you pick, you might have to do additional steps (as shown under WDS)

Note on services

Uninstalling roles/features will also remove any services that came with the installation

Examples ([role] → [respective service]):

RIP Listener feature → Rip Listener service

Telnet Server feature → Telnet service

Internet Information Services (World Wide Web Services) role → World Wide Web Publishing Service

Internet Information Services (FTP Server) role → Microsoft FTP Service

Difference between configuring service and role

Disabling and stopping a service will effectively stop communications and functionalities that the service provides

However, this is a temporary action - someone can re enable the service.

Uninstalling the role/feature that the service came with will ensure that the service is not accidentally/intentionally going to be turned on

- Roles/features management requires administrative privileges, whereas services management doesn’t always